

phase3 | 320 training pts

Velocity RMSE (absolute | relative to truth-RMS):

|                    |          |               |                |
|--------------------|----------|---------------|----------------|
| 1. whole domain    | : 1.171  | (rel 0.1523)  | [122600 cells] |
| 2. thick cylinder  | : 0.6761 | (rel 0.08794) | [15000 pts]    |
| 3. on the cylinder | : 0.5871 | (rel 0.07638) | [6000 pts]     |

Pressure RMSE : 12.7

Momentum-integral force:

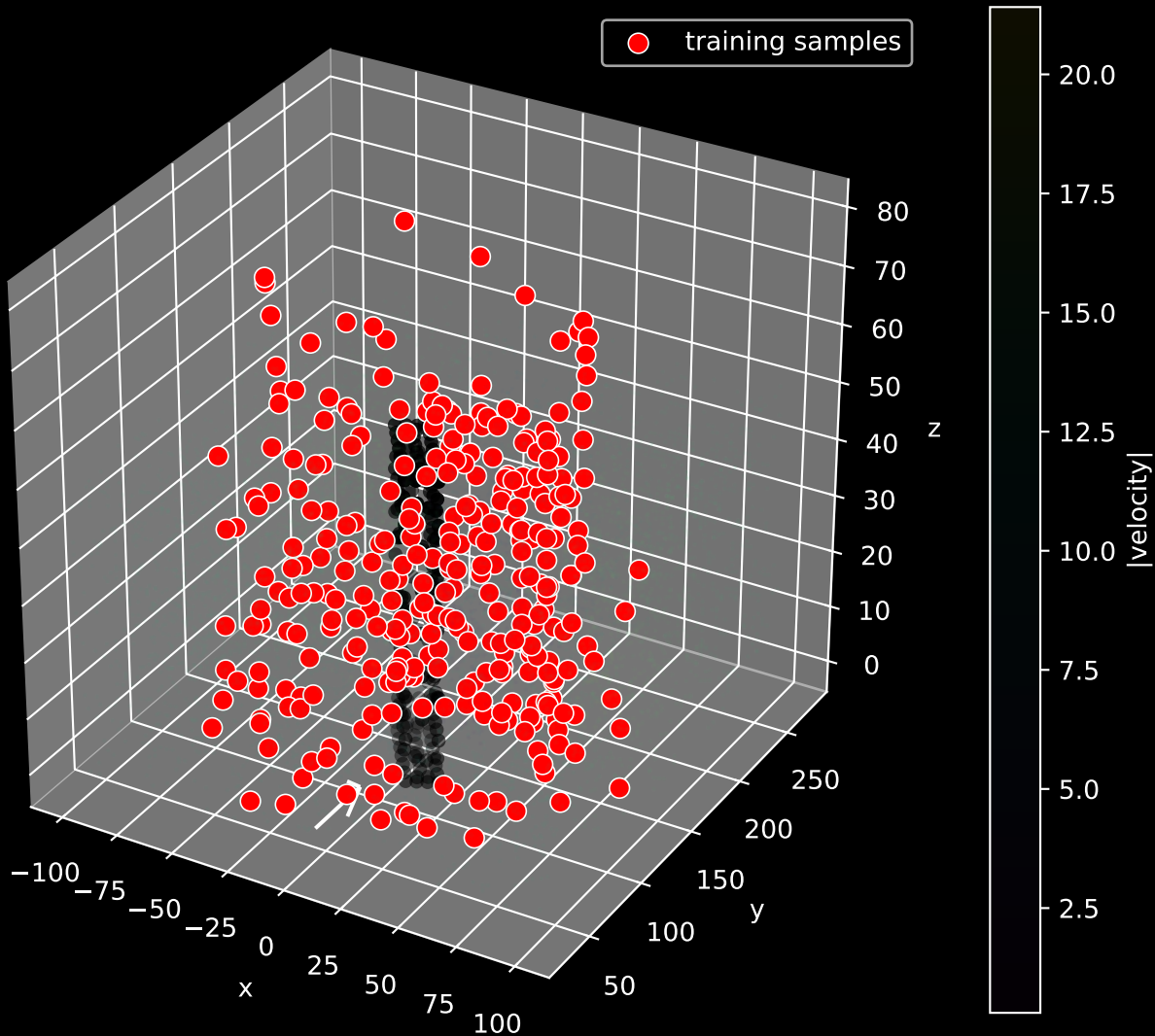
|                |                                     |                     |                |
|----------------|-------------------------------------|---------------------|----------------|
| F              | : 3.179e+05                         | [24945 surface pts] |                |
| F vector       | : [1.62e+05, 2.055e+05, 1.805e+05]  |                     |                |
| F true         | : [1.554e+05, 2.086e+05, 7.259e+04] |                     | F_true =270114 |
| force error dF | : [6545, -3150, 1.079e+05]          | rel=0.4004          |                |

Prior (potential-flow baseline) for reference:

|                    |         |
|--------------------|---------|
| whole domain prior | : 2.953 |
| thick cyl prior    | : 3.32  |
| on cylinder prior  | : 3.266 |

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velocity RMSE 1) whole domain: 1.171 (rel 0.1523) 2) thick cylinder: 0.6761 (rel 0.08794) 3) on cylinder: 0.5871 (rel 0.07638)  
pressure RMSE: 12.7 momentum force |F|: 3.179e+05

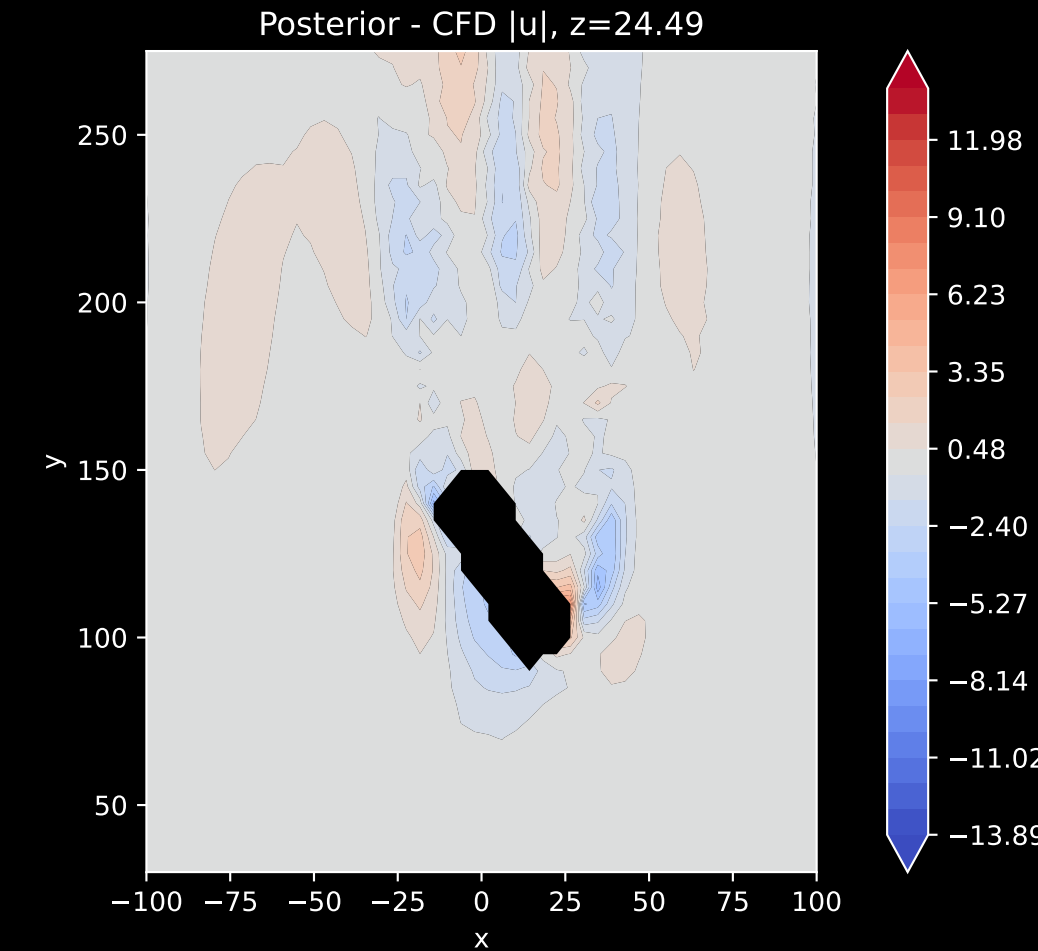
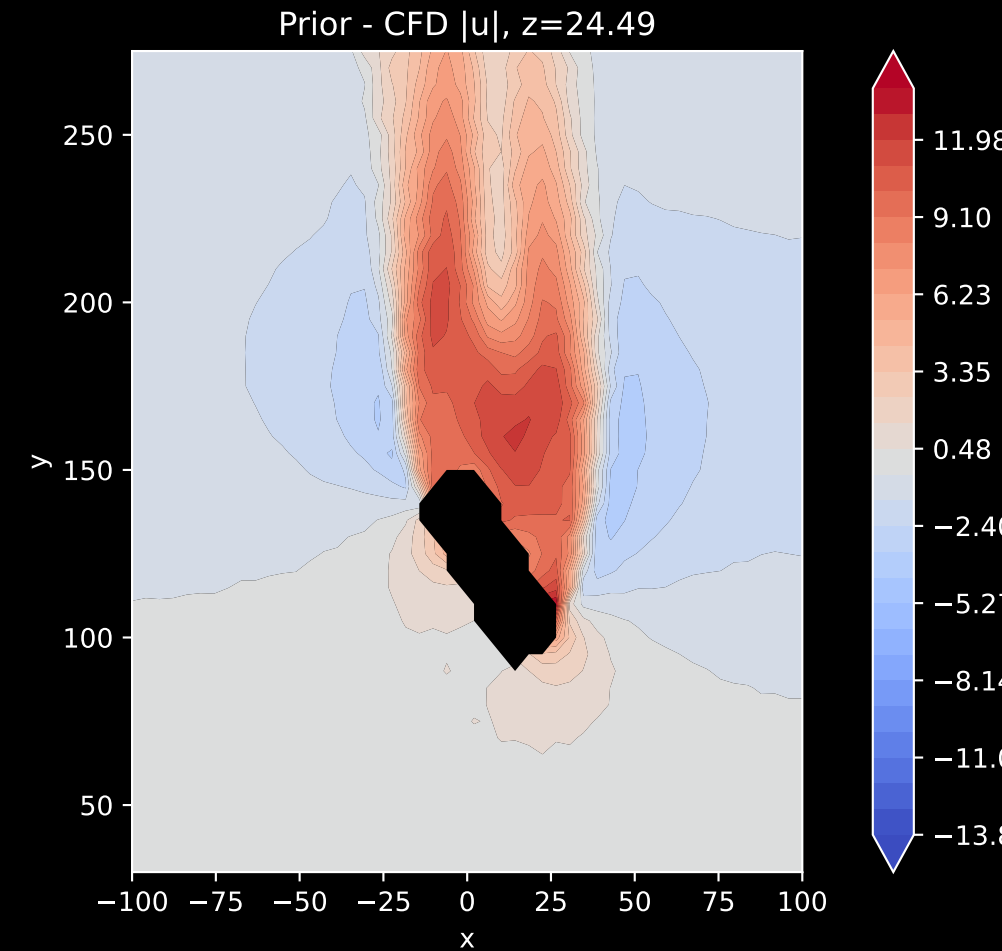
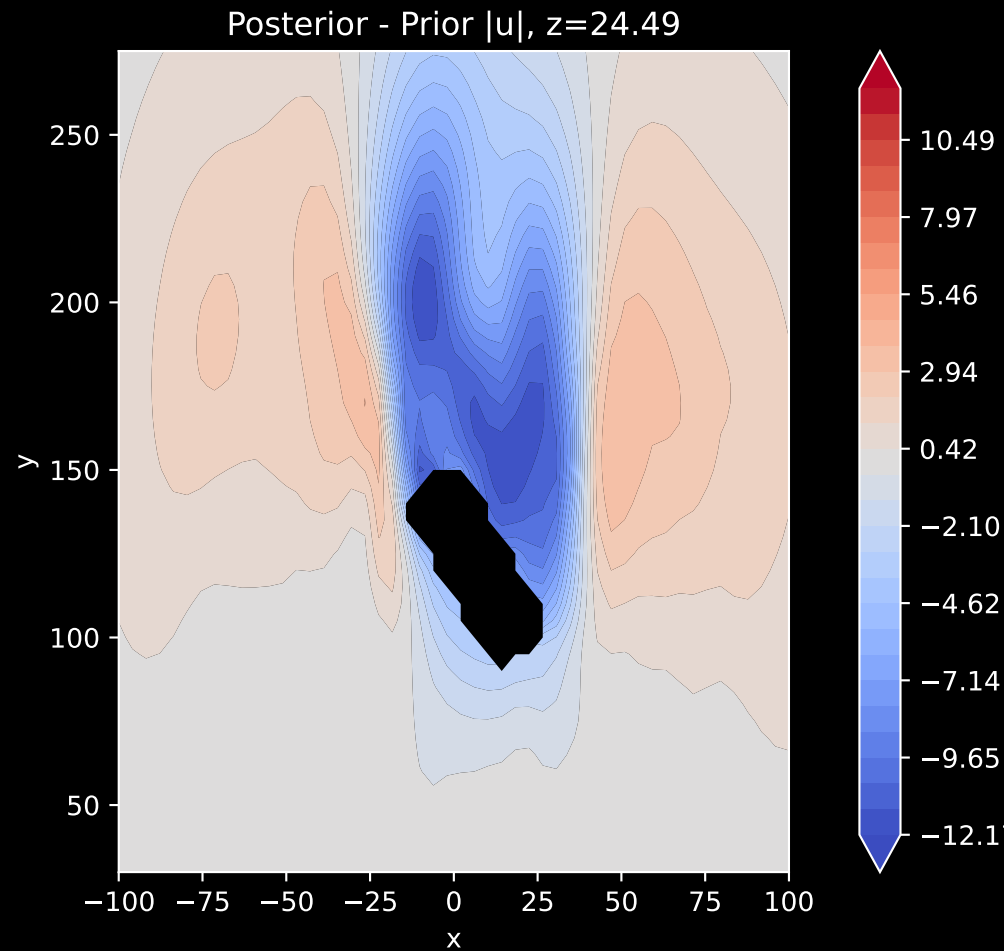
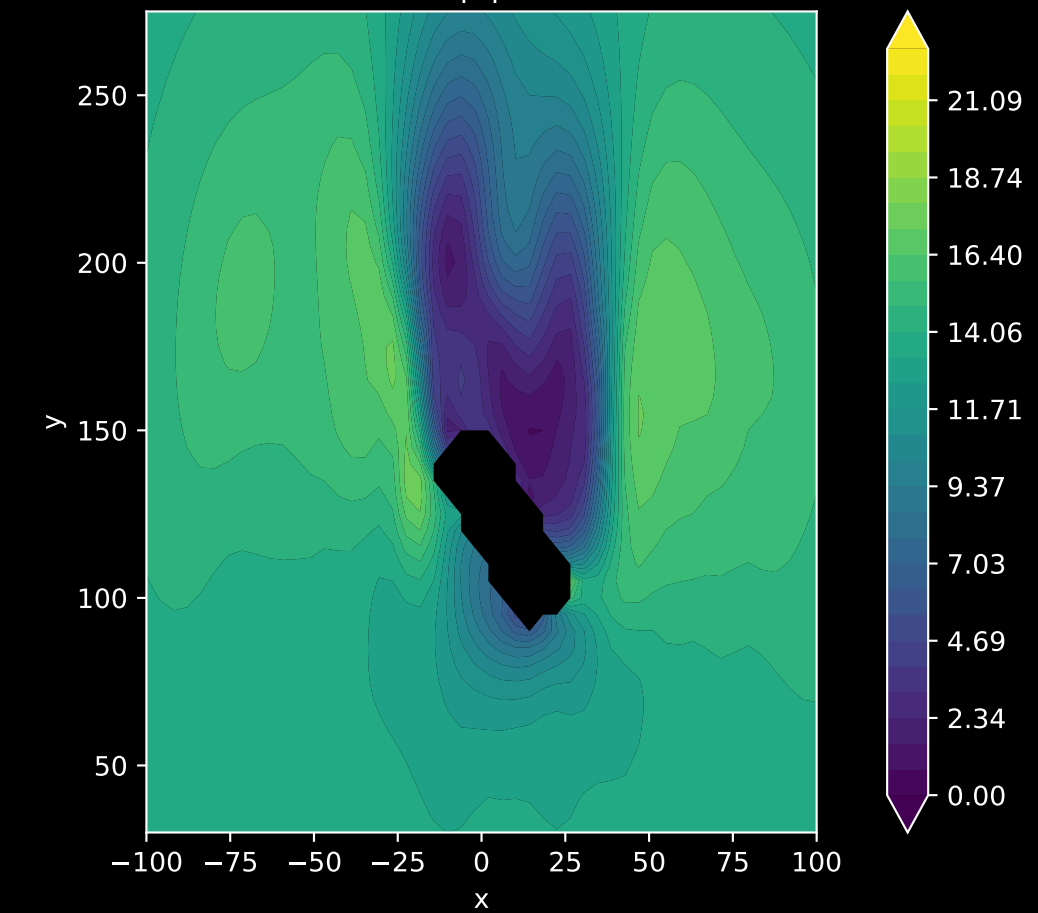
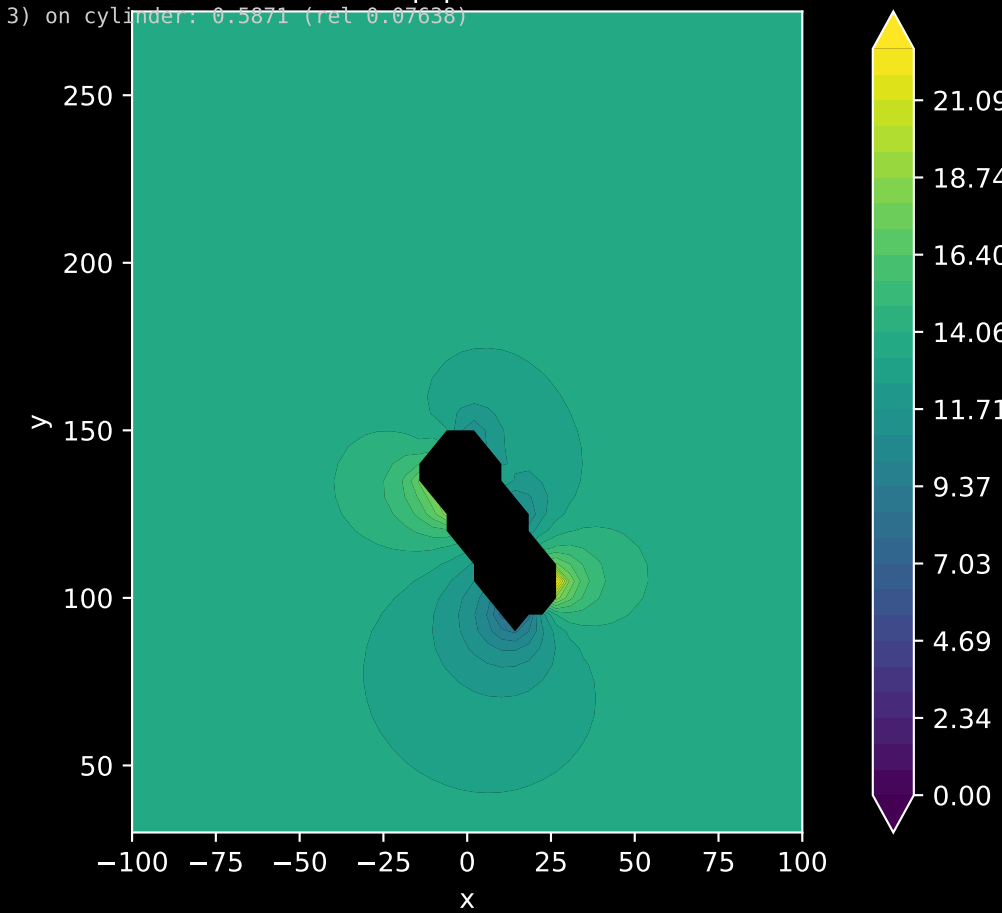
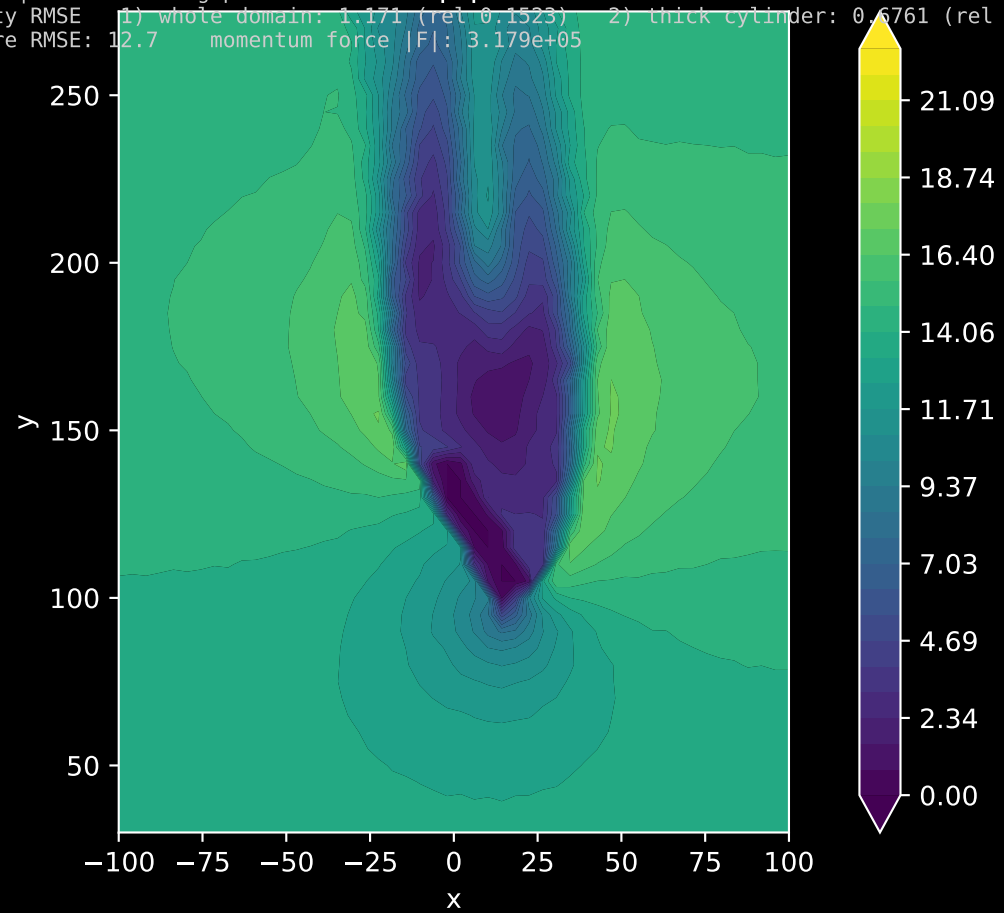
GPR posterior velocity magnitude



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velocity RMSE: 12.7  
pressure RMSE: 12.7

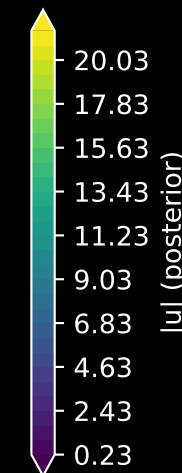
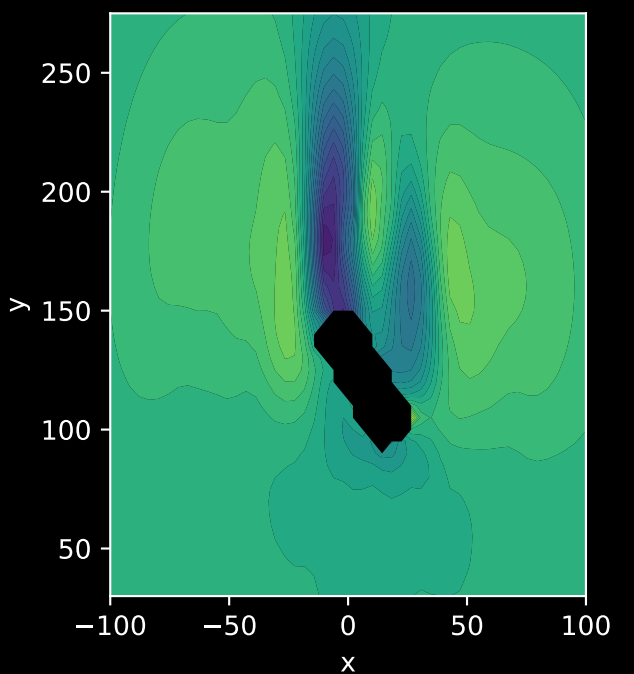
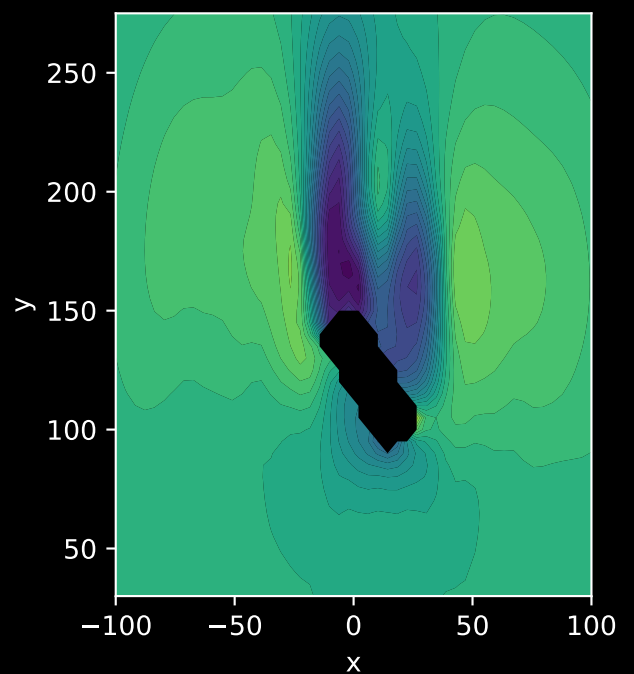
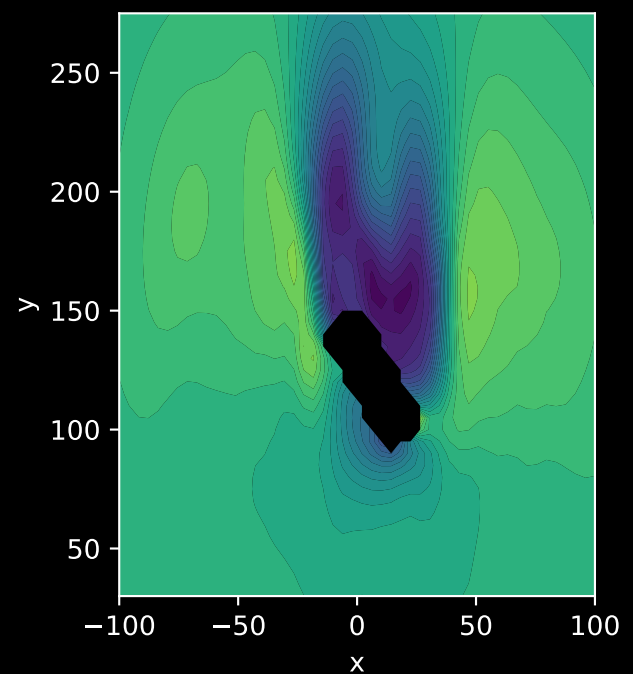
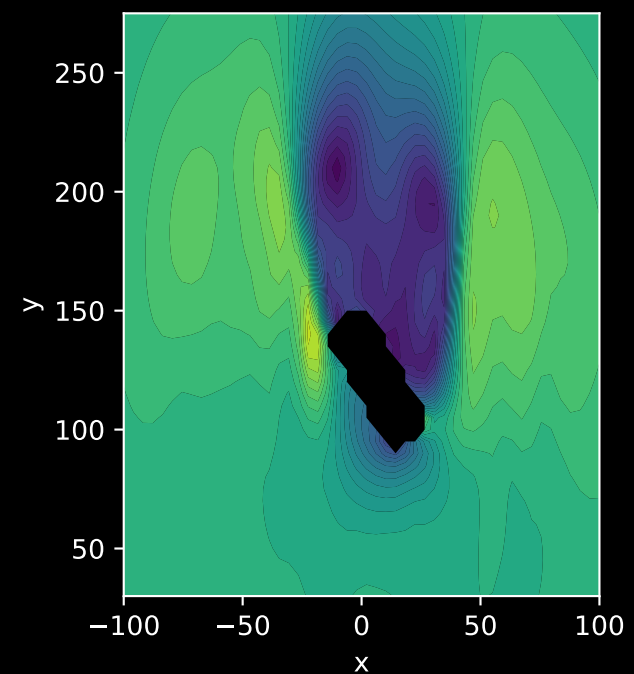
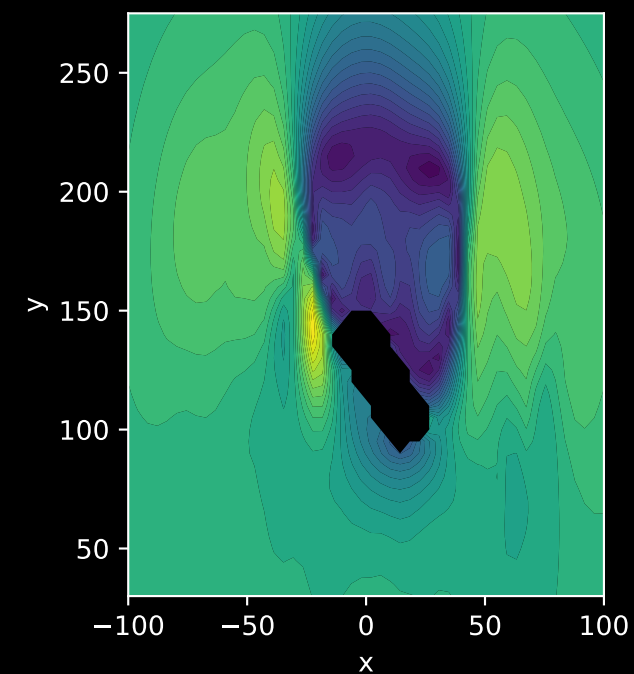
CFD truth  $|u|$ ,  $z=24.49$   
1) whole domain: 1.171 (rel 0.1523)  
2) thick cylinder: 0.6761 (rel 0.08794)  
3) on cylinder: 0.5871 (rel 0.07638)

momentum force  $|F|$ :  $3.179e+05$



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velocity RMSE: 1) whole domain: 1.171 (rel 0.1523) 2) thick cylinder: 0.6761 (rel 0.08794) 3) on cylinder: 0.5871 (rel 0.07638)  
pressure RMSE: 12.7 momentum force |F|: 3.179e+05

posterior  $|u|$  slices along  $z$



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pressure RMSE: 12.7 momentum force |F|: 3.179e+05

variances  $|u|$  slices along z

